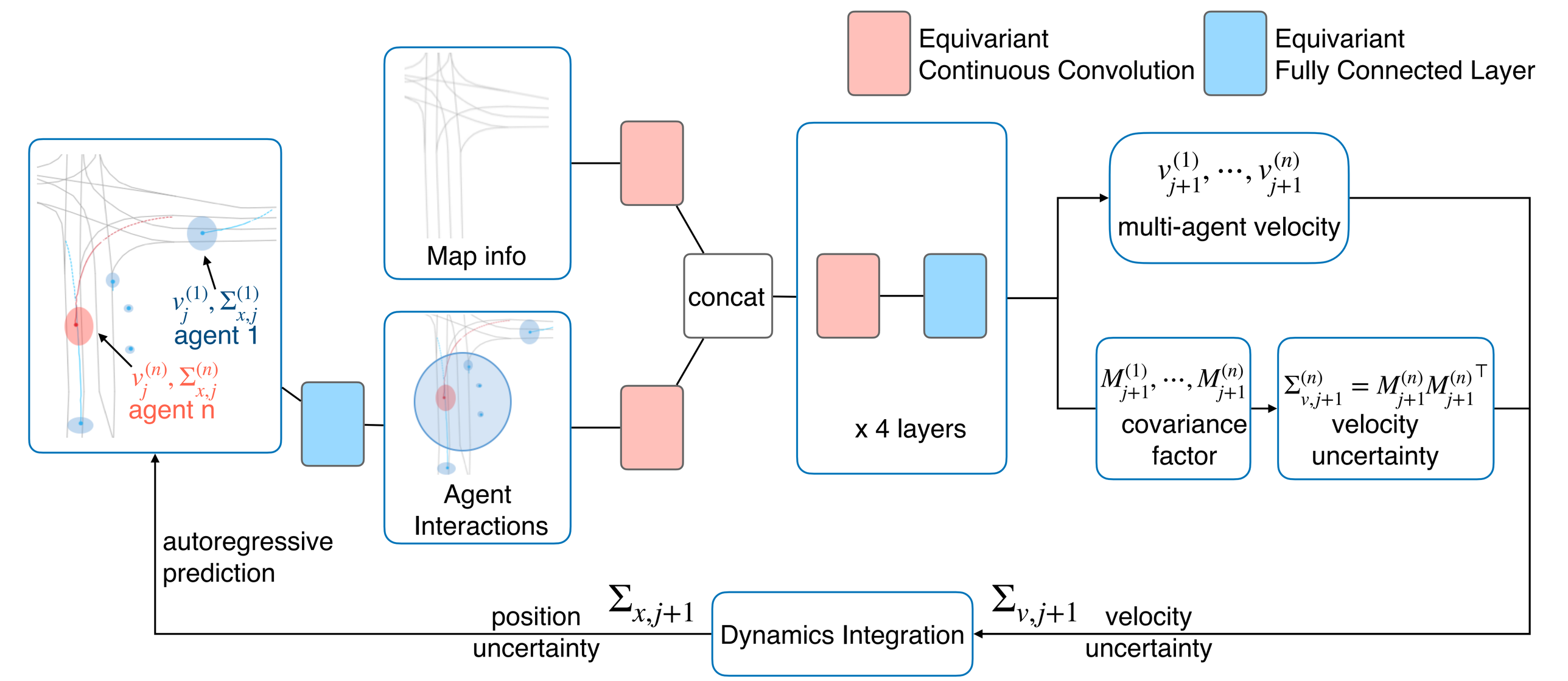
Equivariant
Continuous Convolution
Equivariant
Fully Connected Layer
Map info
$v_j^{(1)}, \Sigma_{x,j}^{(1)}$
agent 1
$v_j^{(n)}, \Sigma_{x,j}^{(n)}$
agent n
Agent
Interactions
concat
x 4 layers
$v_{j+1}^{(1)}, \cdots, v_{j+1}^{(n)}$
multi-agent velocity
$M_{j+1}^{(1)}, \cdots, M_{j+1}^{(n)}$
covariance
factor
$\Sigma_{v,j+1}^{(n)} = M_{j+1}^{(n)} M_{j+1}^{(n)\top}$
velocity
uncertainty
autoregressive
prediction
position
uncertainty
$\Sigma_{x,j+1}$
Dynamics Integration
$\Sigma_{v,j+1}$
velocity
uncertainty